

Cite this: *J. Mater. Chem. C*,
2024, 12, 6106

A terahertz metasurface biosensor based on electromagnetically induced transparency for fingerprint trace detection†

Rong Zhao,^{‡a} Qiang Niu,^{‡a} Yi Zou,^a Bin Cui^{ab} and Yuping Yang^{ID} *^{ab}

A highly sensitive and accurate detection is long-awaited for terahertz (THz) metasurface biosensors through a comprehensive multi-modal sensing and fingerprint trace enhancement. Herein, we present a polarization-insensitive THz biosensor with a sensitivity up to 411.6 GHz per RIU (4- μ m-thickness analyte) based on an electromagnetically induced transparency (EIT) metasurface. Moreover, the carcinoembryonic antigen is accurately identified through a multi-modal resonance, achieving a limit of detection (LOD) of 2.5 ng, and a qualitative and quantitative investigation into an enhanced fingerprint detection for extremely thin analytes is also conducted, which reveals that the transparency peak is exceptionally sensitive to the fingerprint vibration of substances through its amplitude variation. Thus, the proposed THz EIT metasurface achieves a highly sensitive sensing and fingerprint detection, which is beneficial for qualitative and quantitative biological identifications in the fields of biomedicine, food safety and drug detection to some extent.

Received 2nd February 2024,
Accepted 28th March 2024

DOI: 10.1039/d4tc00455h

rsc.li/materials-c

1. Introduction

Metasurface, exciting various resonance modes, can considerably enhance the interaction between light and matter by artificially manipulating electromagnetic characteristics.^{1–5} The frequency shift or amplitude change of resonant peaks or dips has become an effective method to achieve qualitative or quantitative biosensing of trace analytes. Up to now, a variety of methods have been attempted to reach a greater frequency shift or higher *Q*-factor to improve the sensitivity of meta-sensing.^{6,7} Recently, electromagnetically induced transparency (EIT) arising from quantum interference in three-level atomic systems has been extended to metasurfaces for the realizations of slow-light devices,^{8,9} absorbers,¹⁰ modulators,¹¹ and biosensors.^{12,13} In 2008, Zhang *et al.*⁹ demonstrated a plasmonic metamaterial to investigate theoretically EIT phenomena resulting from a radiative element coupled with a subradiant (dark) element. Gu *et al.*¹¹ achieved an active optical control of EIT metasurface by integrating photoconductive silicon. Kang *et al.*¹² proposed a T-shaped EIT resonator, which was attributed to the coupling between bright and dark modes as well as to the structural-symmetry breaking. More

recently, EIT metamaterials are especially attractive for biosensing applications because of their multiple resonance modes, including two absorption dips and one transparency peak. In 2019, Yan *et al.*¹³ presented a symmetry-breaking double SRR (split-ring resonator) excited by the EIT effect for a high sensitivity detection of cancer cells and anti-cancer drug concentrations. Specifically, the sharp transparency peak within a broad absorption band exhibits an exceptionally high-sensitivity response to the surrounding dielectric environment and shows enormous application potential in biosensing.¹⁴ However, few studies have focused on sensing applications utilizing the transparency peak in the terahertz (THz) range, and the integrated verification platform that incorporates amplitude and resonant frequency.

The THz spectroscopy technique can reveal inter- or intramolecular vibrational modes, providing distinctive fingerprint information with label-free and non-invasive detection.^{15–17} The length mismatch between the scales of molecules and THz wavelengths has been gradually mitigated due to the localized field enhancement induced by surface plasmon resonance (SPR) of metasurfaces. Furthermore, the advancement of pixelated construction and angle-multiplexed technology^{18,19} has significantly highlighted the THz fingerprint response to substances through the amplitude variation of SPR peaks or dips. Nevertheless, challenges persist in maintaining amplitude stability, spread spectrum capabilities, and high sensitivity for analytes using these techniques. Hence, there is an urgent demand to introduce an enhanced fingerprint signal to highlight the characteristics of THz spectroscopy.

^a School of Science, Minzu University of China, Beijing 100081, China.

E-mail: ypyang@muc.edu.cn

^b Engineering Research Center of Photonic Design Software, Ministry of Education, Beijing 100081, China† Electronic supplementary information (ESI) available: See Supplement 1. See DOI: <https://doi.org/10.1039/d4tc00455h>

‡ These authors contributed equally to this work.

In this work, a high-sensitivity EIT metasurface is proposed to demonstrate a multi-modal THz biosensing and an enhanced fingerprint detection, which further promotes the sensing applications of transparent peak characteristics. Three primary aspects are covered as follows: (1) the evolution mechanism of the EIT metasurface is discussed and explained by theoretical calculations and numerical simulations; (2) a novel scheme combining frequency shift and amplitude variation is introduced to achieve a high-accuracy mutual verification detection, subsequently enabling multi-modal THz biosensing, and (3) the pixelated broadband constructed by the high-sensitivity response of a transparent peak to vibration absorption was used to obtain a fingerprint enhanced detection at an extreme thickness as thin as 45 nm.

2. Design and methods

The designed metasurface biosensor is illustrated in Fig. 1(a), consisting of a four-split ring (FSR) resonator and a 1-mm-thickness quartz substrate with a relative dielectric constant of 3.69. The geometrical parameters are listed in Table 1, including period (p), outer ring radius (R), inner ring radius (r), line width (w), 8-shaped split ring width (D) and open gap (d). The proposed metasurface biosensor was fabricated by conventional photolithography and metallization processing. First, 50-nm Ti was set as the bonding layer, and 200-nm aluminum was sequentially deposited on the quartz substrate. Then, a lift-off process was performed to obtain the designed meta-biosensor. Fig. 1(b) displays the microscopic image of the fabricated EIT array.

Table 1 Geometric parameters of the unit cell (μm)

p	R	r	w	D	d
50	23	20	3	22	3

To gain an insight into sensing mechanisms, finite element analysis (FEA) was used to simulate the sensing performance of the designed EIT metasurface biosensor. In the measurement, a homemade THz time-domain spectroscopy (THz-TDS) system with a frequency range of 0.1–3 THz was used to obtain THz temporal waveforms and transmission spectra.^{20,21} Here, THz signals through the blank quartz substrate and metasurface biosensor with or without analyte were measured to serve as reference and sample signals, respectively. Then, the amplitude transmission was given by $T(\omega) = E_{\text{sam}}(\omega)/E_{\text{ref}}(\omega)$, where $E_{\text{sam}}(\omega)$ and $E_{\text{ref}}(\omega)$ are corresponding fast Fourier transform (FFT) spectra of the time-domain sample and reference signals, respectively. All tests were performed at a temperature of 22 °C and a relative humidity (RH) below 5%.

3. Results and discussion

3.1. Analysis of the theoretical mechanism

To elucidate the underlying resonant mechanism of the proposed EIT metasurface, two additional elements were further fabricated and measured, as shown in Fig. 1(c) and (d). It can be seen that vertical and horizontal 8-shaped split rings were effectively excited by incident THz waves and induced electric- and toroidal-dipole resonances,² respectively, as shown in the inset of Fig. 1(c) and (d), which meets the

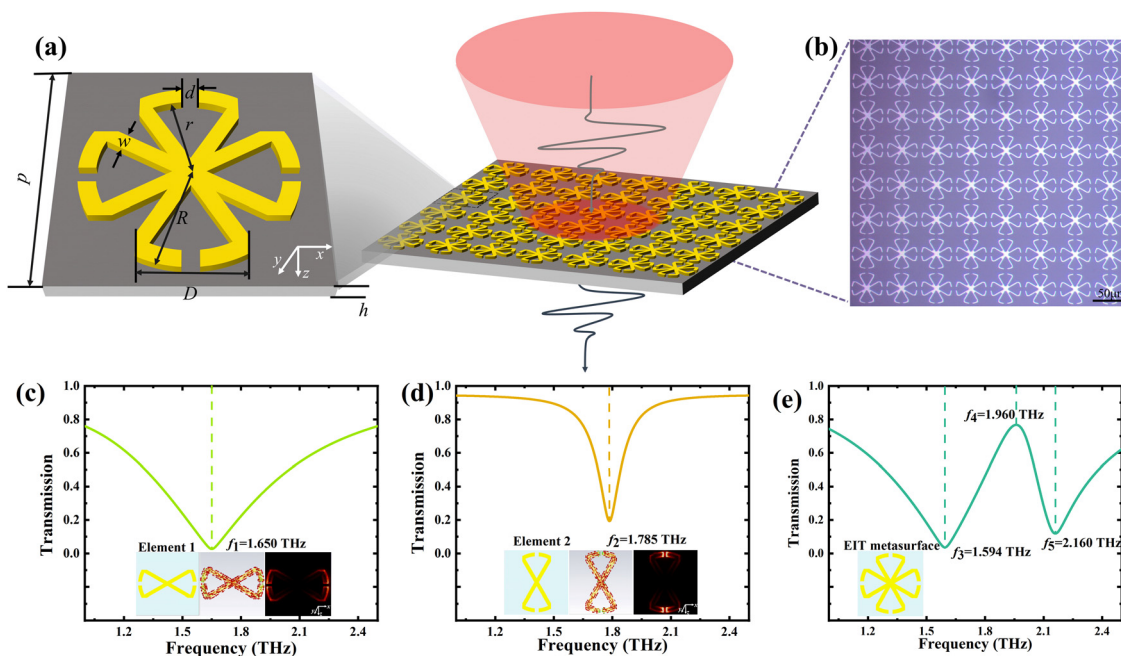


Fig. 1 (a) Schematic configuration of the designed EIT metasurface. The incident THz wave is polarized along the x-axis; (b) microscope image of the fabricated EIT array; transmission spectra of (c) element 1, (d) element 2 as well as (e) the EIT metasurface. The insets show structures, the surface current, and the electric field energy density.

conditions for stimulating the EIT-like phenomenon, resulting in a narrow transparency peak, shown in Fig. 1(e).

Surface current and electric field energy density at Dip 1, Dip 2 and peak are drawn in Fig. S1 (ESI). It can be seen that Dips 1 and 2 are mainly dominated by elements 2 and 1, respectively. In the case of the transparency peak, electric field distribution is highly concentrated at the gaps of element 2 and suppressed at element 1, which is consistent with the three-level system in quantum EIT involving a ground state and two excited states, that is, the electric dipole with a low Q value is directly excited by incident x -polarized terahertz waves and further excites toroidal-dipole resonance by the near field coupling. The destructive interference between toroidal dipole and electric dipole leads to a narrow transparency peak.¹¹ The corresponding dispersion in this frequency range is also greatly changed and leads to a slow light effect, which can be attributed to the sudden phase change caused by the decrease in group velocity, as illustrated in Fig. 2(a). The group delay t_g is defined as $t_g = -d\phi/d\omega$, where ϕ and ω represent the phase and angular frequency, respectively.^{22,23} It can be seen that the group delay effect appears at the narrow transparency peak and reaches 1.273 ps, which is roughly equivalent to an optical distance of 0.382 mm in vacuum.

Fig. S2 1(a) and (b) (ESI†) exhibit transmission spectra and corresponding relative changes in resonant frequency and amplitude caused by rotational perturbations for the proposed metasurface within 45° at steps of 5° to investigate its polarization insensitivity. It can be seen that both signal changes are kept within 2% stemming from the strictly symmetrical structure, which reduces the experimental error arising from polarization deviation to verify the experimental stability and imbues it with the traits of convenience and practicability.

To further explore near-field coupling efficiency between two elements, the Lorentz oscillator model is used to analyze the coupling mechanism between two resonance modes, which can be described as:^{9,11}

$$\begin{aligned} \ddot{x}_1 + \gamma_1 \dot{x}_1 + \omega_0^2 x_1 + \kappa x_2 &= E, \\ \ddot{x}_2 + \gamma_2 \dot{x}_2 + (\omega_0 + \sigma)^2 x_2 + \kappa x_1 &= 0 \end{aligned} \quad (1)$$

where x_1 , x_2 , γ_1 and γ_2 are displacements and damping rates of

two elements, σ denotes the detuning of the resonant frequency of the toroidal-dipole oscillator (element 2) from the electric-dipole one (element 1), κ is the coupling coefficient between two elements, and E is the incident electric field. The transmission t is determined by $t = 1 - g\chi_i$, where g represents the coupling strength of element 1 with E , and χ_i is the imaginary part of susceptibility χ , which can be described as:^{24,25}

$$\chi = \chi_r + i\chi_i \propto \frac{(\omega - \omega_0 - \sigma) + \frac{i\gamma_2}{2}}{\left(\omega - \omega_0 + \frac{i\gamma_1}{2}\right)\left(\omega - \omega_0 - \sigma + \frac{i\gamma_2}{2}\right) - \frac{\kappa^2}{4}} \quad (2)$$

In the proposed EIT scheme, the gap of the FSR resonator (d) emerges as a regulator of the amplitude transmission (Fig. 1(e)), the detuning rate (σ), and the susceptibility (χ). Fig. S3 (ESI†) displays fitting transmission spectra with various d using eqn (2). The dependence of the fitting parameter on d is described in Fig. 2(b). It can be seen that σ increases with the increase of d , indicating that the resonant frequency exhibits an obvious deviation, which is consistent with that in Fig. 5(b). In addition, γ_1 , γ_2 , and κ are nearly constant, showing the stability of the damping rates and coupling coefficient. This will bring a series of benefits that allow for a greater machining fault tolerance as well as consistency in the sensing performance for fingerprint identification.

3.2. Evaluation of sensing performance

To explore the potential sensing capability of the EIT metasurface, a 4- μm -thickness analyte with various refractive indices (n) was used to evaluate the sensitivity (S) by numerical simulations. S is described as $S = \Delta f/\Delta n$, where Δf and Δn are the shift of the resonant frequency and change in the n of the analyte, respectively. A remarkable redshift in Dip 1, Dip 2 as well as the peak was observed upon increasing n from 1.0 to 2.0, as displayed in Fig. 3(a), which is induced by changes in the surrounding dielectric environment. According to previous reports for toroidal- and electric-dipole resonances,^{2,6,7,26} frequency shifts for Dip 1 and Dip 2 are described by $\Delta f_1 \propto 1/2\pi\sqrt{LC}$ and $\Delta f_2 \propto f_0/n_{\text{eff}}$, respectively, where L and C are the equivalent inductance and capacitance; f_0 is the resonant frequency of the bare metasurface, and n_{eff} is the effective refractive index of the environment surrounding the metasurface, depending on the

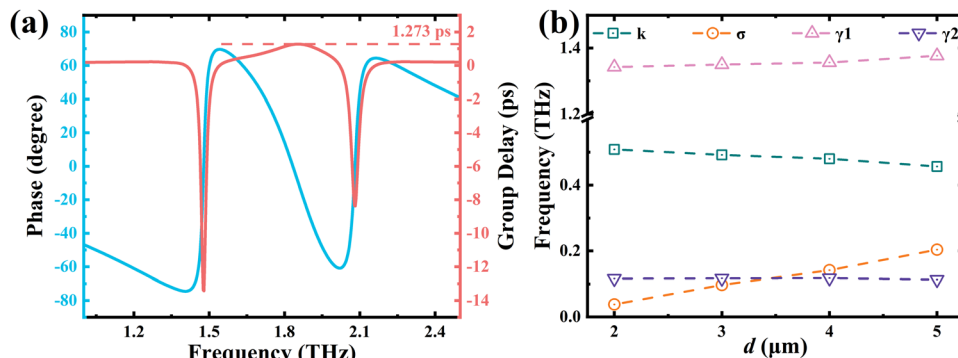


Fig. 2 (a) Phase and group delay for the EIT metasurface; (b) dependence of γ_1 , γ_2 , σ and κ on different d . The unit of κ is THz^2 .

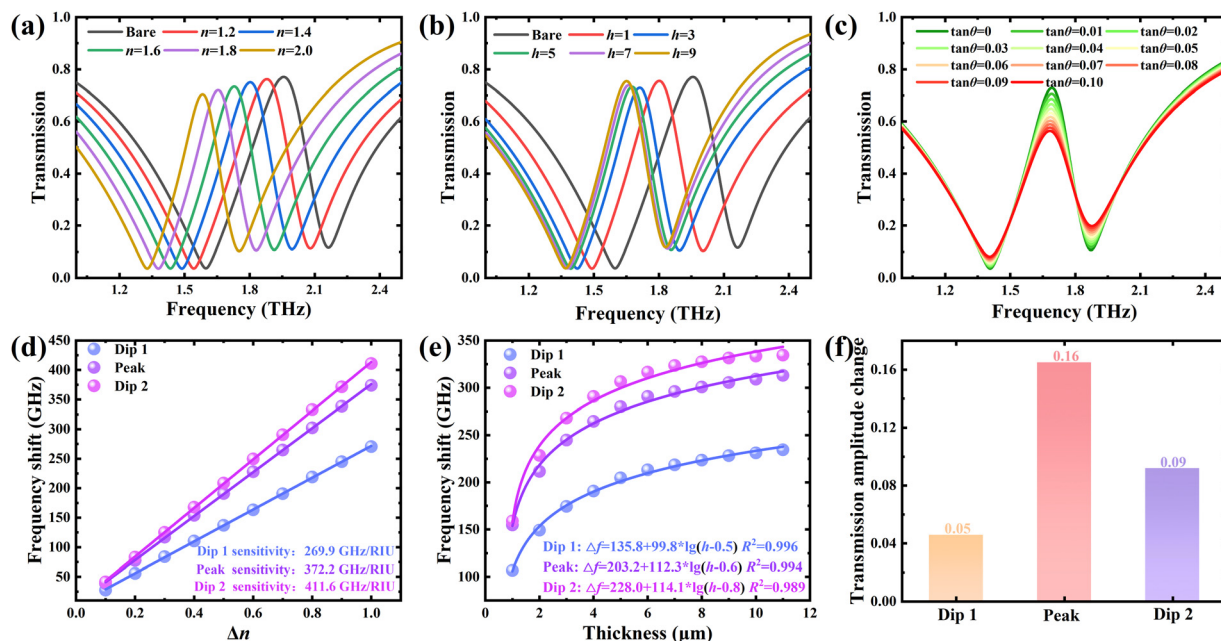


Fig. 3 Transmission spectra for different (a) refractive indices, (b) overlayer thicknesses and (c) $\tan \theta$. (d) Frequency shift (Δf) as a function of the refractive index. (e) Relationship between the frequency shift and thickness. (f) Amplitude changes for Dip 1, peak as well as Dip 2.

refractive index of substrate, analyte and air. C and n_{eff} are effectively changed with the addition of analytes, which further induces a redshift of the resonant frequency. Moreover, the sensing performance for the peak is mainly determined by the coupling between two resonance modes, *i.e.* the electric-dipole mode is suppressed by the toroidal-dipole mode. The relationship between the frequency shift and refractive index can be fitted linearly, where S for Dip 1, Dip 2 as well as peak are theoretically calculated as 269.9, 411.6, and 372.2 GHz per RIU, respectively, as shown in Fig. 3(d). Furthermore, thickness-dependent transmission spectra were investigated to analyze the influence of substance content on the sensing. It can be seen in Fig. 3(b) that the resonance frequency appears as a red-shift trend with increasing thickness, along with a slowdown in the increment rate and reaches saturation at 10 μm with no further appreciable frequency shift,²⁷ which also a high-sensitivity in trace substances. The responded relationship between thickness and frequency shift is recorded and fitted nonlinearly, as shown in Fig. 3(e). All in all, main sensing parameters are summarized in Table 2 for quantitative comparison. Notably, the normalized sensitivity (S/hf_0) is introduced to diminish the influence of analyte thickness and resonant frequency and compared to previously reported biosensors.

Recently, the trace sensing of THz molecular fingerprints has attracted extensive attention, and fingerprint sensing is sensitive to the extinction coefficient of the analyte.¹⁸ Therefore, the relationship between the extinction coefficient and amplitude transmission of the analyte was further studied. By employing the propagation formula of light in a lossy medium, the loss tangent ($\tan \theta = \frac{\epsilon_i}{\epsilon_r}$) is used to analyze the influence of the extinction coefficient (n_i). Fig. 3(c) displays amplitude

Table 2 Parameters of reported metasurface biosensors for comparison

Resonant mode	S (GHz per RIU)	h (μm)	f_0 (THz)	S/hf_0	Ref.
LC	121	5	0.974	24.846	28
LC	135	10	0.835	16.168	29
Fano	165	6	1.17	23.504	30
Toroidal	287.8	4	1.76	40.881	2
Toroidal	403.7	20	1.939	10.410	31
Toroidal	485.3	12	2.39	16.921	32
Toroidal	129.3	28	0.55	8.396	33
Toroidal	27.3	4	0.402	16.978	34
EIT	455.71	11	1.71	24.227	12
EIT	496	10	2.240	22.143	14
EIT	269.9 (Dip 1)	4	1.598	42.225	Our work
	372.2 (peak)	4	1.957	47.547	
	411.6 (Dip 2)	4	2.163	47.573	

LC: inductor-capacitor.

transmission spectra with various $\tan \theta$ (from 0.01 to 0.1). A remarkable decrease in amplitude was observed for the peak compared to those for Dip 1 and Dip 2, revealing that the amplitude of the peak is more sensitive to the extinction coefficient. More importantly, this enables the amplification effect of the strong coupling between transparency peak and molecular vibration, providing high-sensitivity optical probes for material fingerprint identification. In addition, the amplitude changes are recorded and summarized in Fig. 3(f) for comparison. Thus, the transparency peak is more sensitive to the interaction between light and matter, leading to variations in amplitude and position, which provides high sensitivity and multi-mode biosensing.

To validate the high-sensitivity performance of the proposed EIT metasurface biosensor, carcinoembryonic antigen (CEA, a

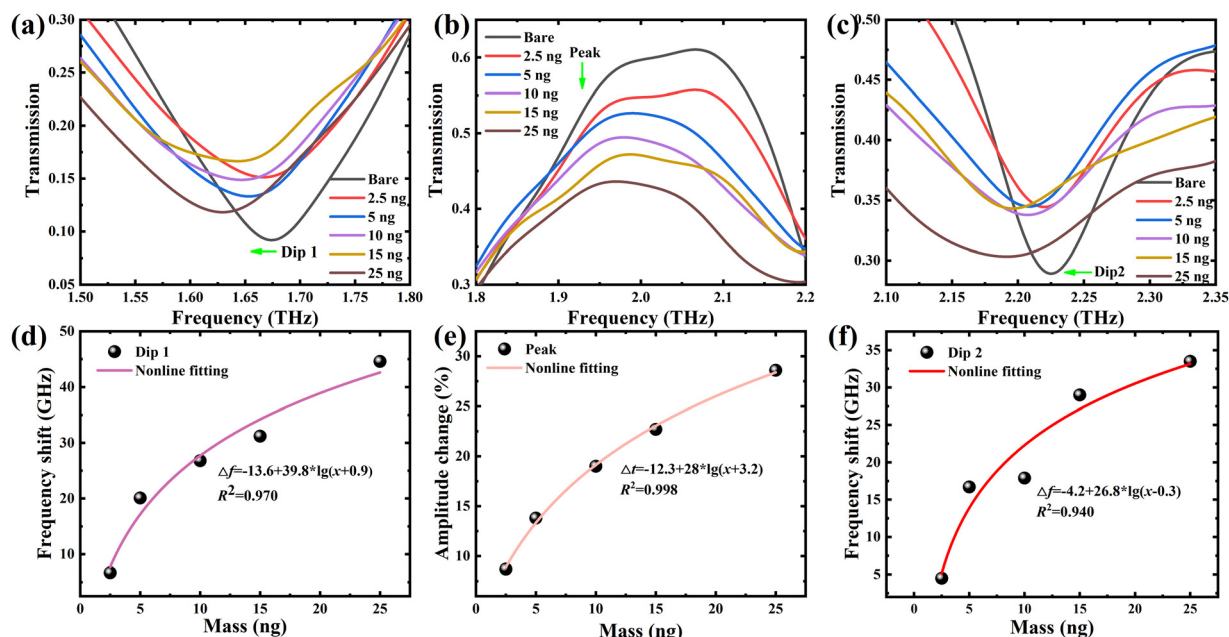


Fig. 4 Transmission spectra of CEA with various contents: (a) Dip 1, (b) peak, and (c) Dip 2; and frequency shift or amplitude for different contents of CEA: (a) Dip 1, (b) peak, and (c) Dip 2.

biomarker highly expressed in patients with cancer) with various contents was deposited on the surface of the proposed EIT biosensor, and the sensing measurement carried out using the THz-TDS system. The measured transmission spectra of Dip 1, peak as well as Dip 2 are illustrated in Fig. 4(a)–(c), respectively. A remarkable frequency red-shift for Dips, as well as an offset in amplitude for the peak, was observed as the CEA content increased from 2.5 to 25 ng. It is worth noting that significant amplitude changes for the transparency peak

appear in a regular evolution, and the frequency shifts of 20.1 and 13.8 GHz for Dip 1 and 2 were observed for 5 ng CEA compared to bare metasurface biosensors, respectively. In addition, the corresponding nonlinear relationship between the frequency shift (Δf) (or amplitude change Δt) and CEA contents (x) was further evaluated and fitted for Dip 1, peak as well as Dip 2, which are described as $\Delta f = -13.6 + 39.8 \times \lg(x + 0.9)$, $\Delta t = -12.3 + 28 \times \lg(x + 3.2)$ and $\Delta f = -4.2 + 26.8 \times \lg(x - 0.3)$, respectively, as shown in Fig. 4(d)–(f). Among them, the amplitude

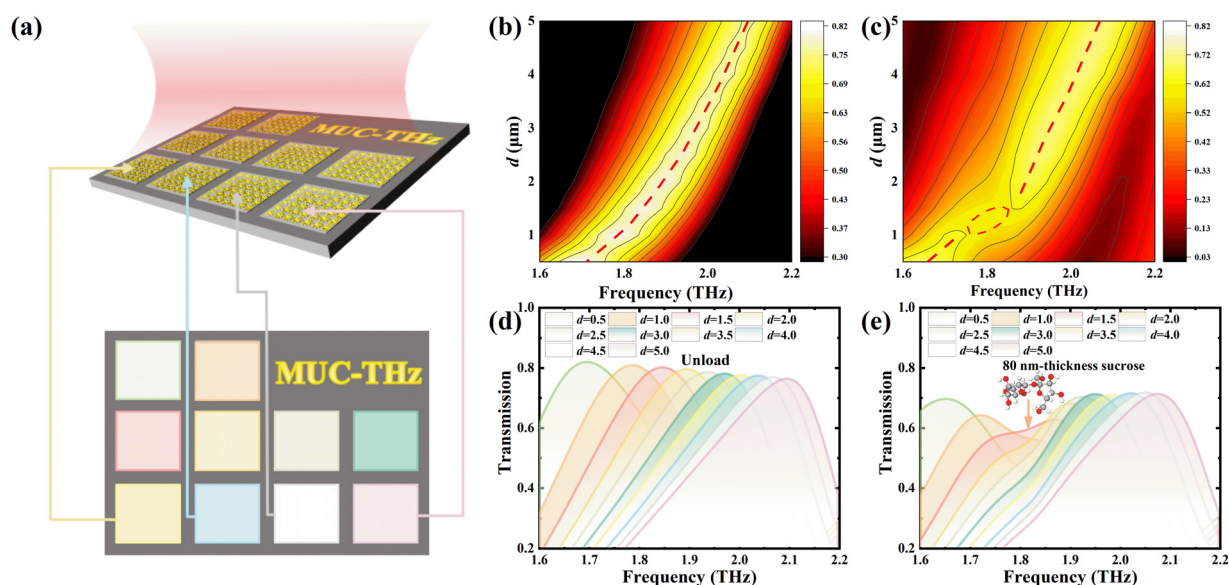


Fig. 5 (a) Schematic representation of the proposed two-dimensional (2D) array of EIT metasurface biosensor pixels; (b) and (c) transmission spectra of 2D EIT arrays with and without a sucrose coating (80 nm thickness); and (d) and (e) transmission spectra of the EIT metasurface with and without a sucrose coating (80 nm thickness) for different d (the unit is μm).

fitting of the transparency peak has the highest identical degree (R^2), indicating a higher quantitative detection accuracy. Due to the multi-modal resonance (frequency shift and amplitude change) provided by the EIT-like mode, a potential approach of accurate identification was verified and achieved. Thus, the proposed EIT metasurface biosensors can be used to achieve high-sensitivity, high-accuracy, and multi-modal sensing characteristics with a limit of detection (LOD) of 2.5 ng for CEA, which is desirable under further extensibility.

3.3. Investigation of fingerprint detection

To comprehensively investigate the sensing performance of the transparency to extinction coefficient and advance the application of EIT metasurface biosensors for the fingerprint detection of trace analytes, we have developed a two-dimensional (2D) array of EIT metasurface biosensor pixels, as illustrated in Fig. 5(a), consisting of ten resonators, which was achieved by adjusting the interference cancellation of two elements through variations in the parameter “ d ” (ranging from 0.5 to 5 μm). Each pixel cell in Fig. 5(a) corresponds to transmission curves with the corresponding color in Fig. 5(d). The pixelated metasurface effectively manipulates resonant frequencies, enabling broadband spectroscopy to simultaneously meet both key criteria for identifying material fingerprints (sensitivity and bandwidth), further facilitating the fingerprinting of multiple analytes. The coupling between metasurface resonances and molecular vibrations allows for the tailoring of the spectral response through molecular absorption. Specifically, we have

observed a significant amplitude dip at a fixed frequency, indicating a resonance enhanced fingerprint absorption of the molecules, and the absorption intensity is proportional to the concentration of the analytes. The relationship provides a robust basis for the identification and quantification of molecular fingerprints. As d increases, the resonance frequency for each pixelated blank metasurface continuously shifts towards higher frequencies, while the line shape remains almost unchanged, as shown in Fig. 5(b) and (d), indicating a uniform coupling performance in a broad spectrum range. For comparison, a sucrose overlayer (80 nm thickness) was deposited on the surface of the pixelated metasurface, which exhibits a discontinuous red-shift in the resonance frequency, and the asymmetry of the line-shape undergoes significant changes, as displayed in Fig. 5(c) and (e). More importantly, a splitting point at approximately 1.82 THz can be remarkably observed from the envelope curve of multiple pixelated metasurfaces, which corresponds to the absorption peak of sucrose,³⁵ as shown in Fig. 5(e). In addition, the fingerprint enhancement effect is determined by the matching of molecule vibrational absorption and metasurface plasmonic resonance. Thus, the coupling interaction between sucrose and each pixelated metasurface can be identified by the strength of the splitting point, which is intuitively demonstrated from the corresponding spectra discrepancy between Fig. 5(d) and (e).

To further explore the thickness-dependence on the fingerprint signal for different coupled strengths, Fig. 6(a)–(c) shows the transmission spectra of coated sucrose layers with different

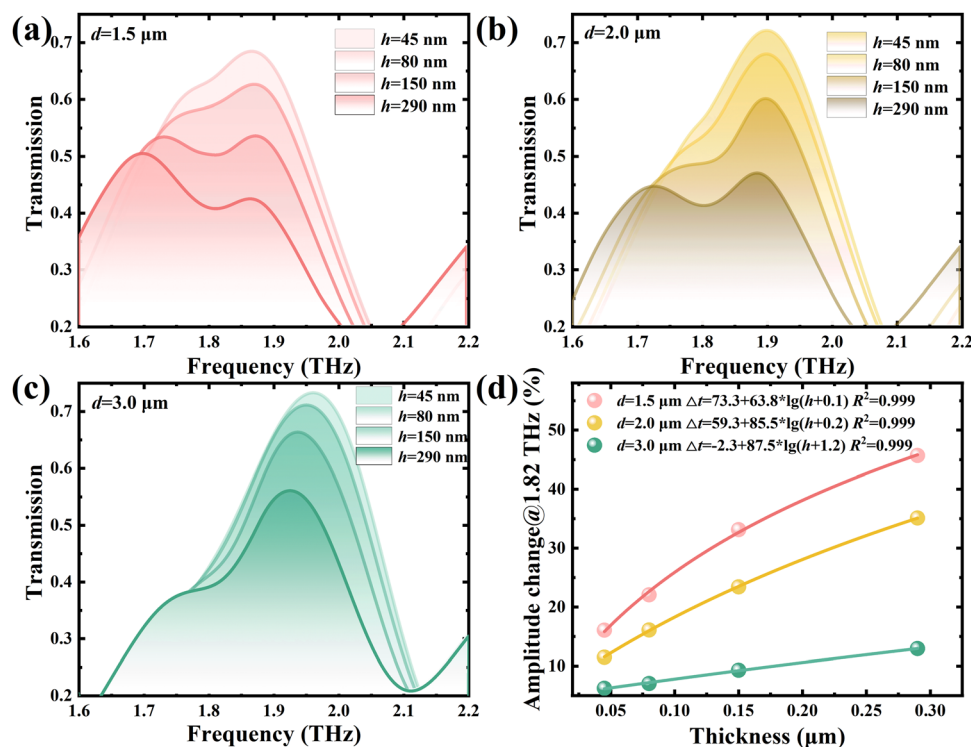


Fig. 6 Transmission spectra of the sucrose layer with different thicknesses (from 45 to 290 nm): (a) $d = 1.5 \mu\text{m}$, (b) $2 \mu\text{m}$, (c) $3 \mu\text{m}$; and (d) the relationship between the amplitude at 1.82 THz and analyte thickness.

thicknesses (from 45 to 290 nm), where d is 1.5, 2 and 3 μm . Specifically, for the transmission spectra of each pixel unit, the fingerprint feature of sucrose can be distinctly observed as the thickness increases, achieving a LOD of thickness 45 nm. Moreover, the coupling strength between each pixelated metasurface and sucrose absorption is characterized by a frequency component at 1.82 THz, which is most likely to create a well-defined fingerprint signal for $d = 1.5$, caused by the vibration matching between the transparent peak and fingerprint absorption. Furthermore, the corresponding relationship between the fingerprint signal and analyte thickness (h) were further evaluated and fitted, which are described in Fig. 5(d), so as to achieve a quantitative and qualitative detection of trace substances with a LOD of 45 nm thickness and obtain visual proof of the coupling strength. This approach also provides the possibility for the realization of a broad-spectrum fingerprint recognition in the single-pixelated metasurface. Hence, the EIT transparent peak is sensitive to the extinction coefficient and achieves a high sensitivity and specific fingerprint detection of trace substances, aiming at an exceptional potential for qualitative and quantitative identifications in the fields of biomedicine, food safety and drug detection.

4. Conclusion

We demonstrated a multi-modal trace-enhanced fingerprint detection using an EIT metasurface. The proposed four-split metal-resonator is extremely sensitive to changes in the dielectric environment and provides an integrated platform for the resonant frequency and amplitude mutually, resulting in an ultralow LOD with high accuracy. Importantly, the pixelated broadband formed through the transparency peak exhibits a remarkable sensitivity to the molecular vibration of the analyte. This success leads to a greatly enhanced fingerprint detection even for extremely thin thicknesses. Hence, the suggested EIT metasurface showcases a significant potential for qualitative and quantitative detection by leveraging the characteristics of transparent peaks. This advancement is expected to propel the growth of THz EIT devices.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This research was funded by the National Natural Science Foundation of China No. 62075248, the National Key R&D Program of China (No. 2020YFB2009300 and 2017YFB0405400), the International Collaboration Project of the Ministry of Science and Technology of China (Grant No. G2022184001L), the Graduate Research and Practice Projects of Minzu University of China (Grant No. SJCX2023019), and the Entrepreneurship Training Program for Undergraduate Students of Minzu University of China (Grant No. MCYX2023110792). We express our gratitude to Ghulam Murtaza for diligently reviewing the English language in this manuscript.

Notes and references

- W. Liu and Y. S. Kivshar, *Opt. Express*, 2018, **26**, 13085–13105.
- Q. Niu, L. Fu, Y. Zhong, B. Cui, G. Zhang and Y. Yang, *Anal. Chem.*, 2023, **95**, 1123–1131.
- J. Xu, Y. Luo, S. Xiao, F. Kang and K. L. Tsakmakidis, *Adv. Opt. Mater.*, 2023, **11**, 2201836.
- Y. Zhou, P. He, S. Xiao, F. Kang, L. Hong, Y. Shen, Y. Luo and J. Xu, *Opt. Express*, 2022, **30**, 27259–27272.
- J. Xu, Y. You, F. Kang, S. Xiao, L. Hong, Y. Shen, Y. Luo and K. L. Tsakmakidis, *arXiv*, 2023, preprint, arXiv:2310.06878, DOI: [10.48550/arXiv.2310.06878](https://doi.org/10.48550/arXiv.2310.06878).
- J. F. O'Hara, R. Singh, I. Brener, E. Smirnova, J. Han, A. J. Taylor and W. Zhang, *Opt. Express*, 2008, **16**, 1786–1795.
- S. Karmakar, D. Kumar, R. K. Varshney and D. R. Chowdhury, *J. Phys. D: Appl. Phys.*, 2020, **53**, 415101.
- N. Liu, T. Weiss, M. Mesch, L. Langguth, U. Eigenthaler, M. Hirscher, C. Sönnichsen and H. Giessen, *Nano Lett.*, 2010, **10**, 1103–1107.
- S. Zhang, D. A. Genov, Y. Wang, M. Liu and X. Zhang, *Phys. Rev. Lett.*, 2008, **101**, 047401.
- R. Taubert, M. Hentschel, J. Kästel and H. Giessen, *Nano Lett.*, 2012, **12**, 1367–1371.
- J. Gu, R. Singh, X. Liu, X. Zhang, Y. Ma, S. Zhang, S. A. Maier, Z. Tian, A. K. Azad, H.-T. Chen, A. J. Taylor, J. Han and W. Zhang, *Nat. Commun.*, 2012, **3**, 1151.
- C. Liu, D. Li, P. Boutinaud, J. Xu, Y. Du, Y. Hou, C. Zhang, Q. Zhou and F. Kang, *Adv. Photonics Res.*, 2023, **4**, 2200356.
- X. Yan, M. Yang, Z. Zhang, L. Liang, D. Wei, M. Wang, M. Zhang, T. Wang, L. Liu, J. Xie and J. Yao, *Biosens. Bioelectron.*, 2019, **126**, 485–492.
- J. Zhang, N. Mu, L. Liu, J. Xie, H. Feng, J. Yao, T. Chen and W. Zhu, *Biosens. Bioelectron.*, 2021, **185**, 113241.
- Y. Yang, D. Zhai, Z. Zhang and C. Zhang, *J. Infrared, Millimeter, Terahertz Waves*, 2017, **38**, 1232–1240.
- R. Zhao, B. Zou, G. Zhang, D. Xu and Y. Yang, *J. Phys. D: Appl. Phys.*, 2020, **53**, 195401.
- X. Yang, X. Zhao, K. Yang, Y. Liu, Y. Liu, W. Fu and Y. Luo, *Trends Biotechnol.*, 2016, **34**, 810–824.
- J. Lyu, S. Shen, L. Chen, Y. Zhu and S. Zhuang, *Photonix*, 2023, **4**, 28.
- Y. Xie, X. Liu, J. Zhou, H. Zhang, J.-Y. Lin, W. Chen, L.-G. Zhu, K. Meng, Q. H. Liu and J. Zhu, *IEEE Trans. Microwave Theory Tech.*, 2023, 1–10.
- N. Qi, Z. Zhang, Y. Xiang, Y. Yang, X. Liang, P. de and B. Harrington, *Anal. Methods*, 2015, **7**, 2333–2338.
- Q. Niu, R. Zhang and Y. Yang, *Front. Phys.*, 2022, **10**, 859924.
- M. D. Lukin and A. Imamoglu, *Nature*, 2001, **413**, 273–276.
- C. Luo, F. Guo, W. Wan, Y. Fang, P. Wang and X. Huang, *Opt. Laser Technol.*, 2022, **149**, 107926.
- W. Luo, W. Cai, Y. Xiang, L. Wang, M. Ren, X. Zhang and J. Xu, *Opt. Express*, 2016, **24**, 5784–5793.
- S. Xiao, T. Wang, T. Liu, X. Yan, Z. Li and C. Xu, *Carbon*, 2018, **126**, 271–278.

- 26 S. Banerjee, C. S. Amith, D. Kumar, G. Damarla, A. K. Chaudhary, S. Goel, B. P. Pal and D. R. Chowdhury, *Opt. Commun.*, 2019, **453**, 124366.
- 27 J. F. O'Hara, W. Withayachumnankul and I. Al-Naib, *J. Infrared, Millimeter, Terahertz Waves*, 2012, **33**, 245–291.
- 28 Y. Yang, D. Xu and W. Zhang, *Opt. Express*, 2018, **26**, 31589–31598.
- 29 X. Hou, X. Chen, T. Li, Y. Li, Z. Tian and M. Wang, *Opt. Mater. Express*, 2021, **11**, 2268.
- 30 R. Wang, L. Xu, J. Wang, L. Sun, Y. Jiao, Y. Meng, S. Chen, C. Chang and C. Fan, *Nanoscale*, 2021, **13**, 18467–18472.
- 31 C. Zhang, T. Xue, J. Zhang, Z. Li, L. Liu, J. Xie, J. Yao, G. Wang, X. Ye and W. Zhu, *Biosens. Bioelectron.*, 2022, **214**, 114493.
- 32 C. Zhang, T. Xue, J. Zhang, L. Liu, J. Xie, G. Wang, J. Yao, W. Zhu and X. Ye, *Nanophotonics*, 2022, **11**, 101–109.
- 33 J. Xu, D. Liao, M. Gupta, Y. Zhu, S. Zhuang, R. Singh and L. Chen, *Adv. Opt. Mater.*, 2021, **9**, 2100024.
- 34 M. Gupta, Y. K. Srivastava, M. Manjappa and R. Singh, *Appl. Phys. Lett.*, 2017, **110**, 121108.
- 35 M. Walther, B. M. Fischer and P. Uhd Jepsen, *Chem. Phys.*, 2003, **288**, 261–268.